

SAILAKSHMI B T

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PROFESSIONAL SUMMARY

Robotics and Automation undergraduate (CGPA 8.99) at Madras Institute of Technology with hands-on experience in embedded systems, robotics, and AI-driven applications. Skilled in Arduino-based system design, sensor interfacing, and rapid prototyping. Interests span Computer Vision, industrial automation, embedded intelligence, and intelligent robotic systems.

EDUCATION

Bachelor of Engineering – Robotics and Automation

Madras Institute of Technology | 2024 – 2028 | CGPA: 8.99 / 10.0

- Higher Secondary Certificate (2024) – 94.16%
- Secondary School Certificate (2022) – 95.2%

INTERNSHIP & TRAINING

Computer Vision & AI Summer Intern – PixelBotix

June 2026

- Developed practical skills in Computer Vision, image processing, object detection, and AI-based visual recognition through intensive hands-on training and worked on some basic projects.

Industrial Robotics & Automation Training – Birla Carbon & KUKA

February 2026

- Gained exposure to industrial robotic arm programming, Industry 4.0 concepts, automation workflows, and industrial safety practices.

PROJECTS & RESEARCH

Capstone Project: Student Life Concierge AI Agent (Kaggle & Google 5-Day AI Agents Intensive)

Concierge Agents Track - Submission Grade | Google Agent Development Kit (ADK), Gemini API, Python, FastAPI, Pytest, Uvicorn

- Developed a production-grade, hierarchical multi-agent Capstone project to automate academic workload management by parsing student email databases for deadlines and reserving calendar focus blocks.
- Implemented an input validation and containment layer to protect the agent against adversarial instruction prompts (jailbreak attacks), achieving a 100% security and routing verification score.
- Built the backend using FastAPI and established a complete test suite with Pytest, using mock services to verify performance under local evaluation checks.

Orion Local AI Assistant

Python, FastAPI, WebSockets, Ollama, Llama 3.2, Web Speech API

- Engineered a fully offline, privacy-first AI assistant by integrating Meta's Llama 3.2 with Ollama for local LLM inference, eliminating cloud dependency while preserving low-latency performance.
- Built an asynchronous FastAPI backend with WebSocket communication to deliver real-time agent state transitions (Standby, Thinking, Speaking) and complete, lower-latency local response deliveries.
- Implemented a function-calling (tool-use) framework enabling the AI to autonomously execute backend actions, including live CPU/RAM/disk diagnostics and file read/list operations within a local workspace.
- Designed a modern, glassmorphic web interface with integrated Speech-to-Text (STT) pipelines for hands-free, voice-driven interaction.

Mechanical Rotary Device [Design Patent Pending]

- Designed a dual-mode rotary mechanism integrating a Geneva wheel and dog clutch system using SolidWorks; filed for a design patent.

Grip Stability & Stress Analyzer

- Developed a load cell-based system using HX711 and Arduino Uno to evaluate grip stability and tremor patterns for stress-level estimation.

Adaptive Self-Mutating Authentication System

- Designed an 8051-based digital lock using hardware timers to generate non-deterministic passwords after each login, mitigating replay attacks and shoulder-surfing.

Single-Axis Gimbal System

- Engineered a stabilization gimbal using Arduino Nano, MPU-6050, and SG90 servo motor, achieving real-time angular stability through inertial sensing and closed-loop feedback.

HACKATHONS & COMPETITIONS

Smart India Hackathon 2025 - Finalist

- Contributed to AI/ML-based identification of cryptographic primitives and protocols in multi-architecture firmware binaries.

H2TS Hackathon 2025 - 2nd Place

- Collaborated to develop an AI-driven SOS platform for emergency situations.

VISA 24 Hours Hackathon 2026 - Participant

- Competed in a 24-hour innovation sprint.

TECHNICAL SKILLS

Control systems: PID Control, Motion Control, Stabilization Mechanisms, Closed-Loop Feedback

Programming & Analysis: Python (Intermediate), Embedded C (Fundamentals), MATLAB/Simulink (Intermediate)

Embedded Systems: 8051 Architecture (AT89S52), Arduino Uno R3, Arduino Nano, Sensor Interfacing, Timer-Based Logic

AI & Agentic Systems: Google Agent Development Kit (ADK), Gemini API, Ollama, Llama 3.2, FastAPI, Pytest, Uvicorn, Prompt Engineering

Web & Real-Time Systems: WebSockets, Asynchronous Programming, Speech Recognition & Synthesis (Web Speech API)

Design Tools: SolidWorks, KiCAD, Canva

Cybersecurity & RE: Binwalk, Ghidra, Ubuntu, Firmware Binary Analysis (working knowledge of tools and workflow)

Soft Skills: Communication, Anchoring, Content Creation, Critical Thinking, Problem Solving

CERTIFICATIONS

- AI Capabilities and Limitations – Anthropic
- Foundations of Prompt Engineering – AWS Training & Certifications
- Python for Absolute Beginners – Udemy
- Basics of HTML – Udemy
- Automation in Manufacturing – NPTEL Elite Certification (67%)

LEADERSHIP & EXTRACURRICULARS

Placement Representative	<i>2026 – Present</i>
Serving as liaison between students and placement coordinators for placement-related communications.	
YRC Volunteer	<i>2024 – Present</i>
Helped organize Youth Fest events for visually challenged individuals.	
Member, ARME	<i>2024 – Present</i>
Association of Robotics and Mechatronics Engineers.	
Joint Treasurer, MIT Quill	<i>2025 – 26</i>
Managed treasury operations for the college literary club.	
Office Bearer, SAHA	<i>2025 – 26</i>
Science and Humanities Association — organized and coordinated academic events.	

LANGUAGES

English (Proficient) | Tamil (Proficient) | Telugu (Conversational) | Japanese (Elementary) | Hindi (Elementary)